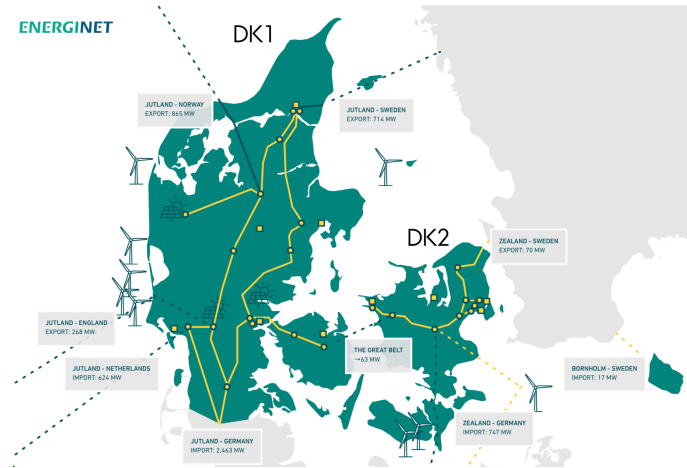


Econometric Game 2026: Electricity Price Zone DK1 Forecasting

Drishti Angra, Clayton Aaron Bytyci, David Chen, and Cai Parry-Jones*

Keywords: Energy & Electricity Markets; LSTM Neural Networks; Panel Model; Ensemble



CONTENTS

Abstract	2
I. Introduction	2
Overview	2
II. Data	3
A. Variable Selection & Feature Engineering	5
B. Exploratory Data Analysis	7
III. Methodology	11
A. Methodology Overview	11
B. Task 1a: Panel Regression	13
C. Task 1b: LSTM Architecture	14
D. Task 2: Calendar Regression	14
IV. Results, Discussion & Robustness	15
A. Task 1: Panel, LSTM and Ensemble	15
B. Task 2: Calendar Regression	16
V. Extensions	19
A. Systematic Variable Selection	19
B. Task 1 – Extension of ensemble with tree based models	19
C. Task 2 – Prediction of Exogenous Variables	19
VI. Conclusion	20
Declarations	20
A. Additional Figures	21
References	22

* All from the Faculty of Economics, University of Cambridge.

ABSTRACT

We use a panel regression model and LSTM neural network model to construct hourly electricity spot price forecasts for the first week of 2024 using a novel subset of variables motivated by findings on data analysis and literature. For our predictions across the first week, we find that a partial ensemble over our models, which exploits their respective strengths, proves to be superior in generating reliable forecasts, exploiting the memory of the LSTM and general dynamics of the panel method used. For our year ahead predictions of hourly spot prices for every week in 2024, we use a calendar model which provides strong forecasts but is vulnerable to systematic seasonal and holiday biases with extensions discussed in this paper.

I. INTRODUCTION

This paper is concerned with forecasting hourly electricity prices in the DK1 price zone: we forecast hourly electricity prices for the first week of 2024 (7×24) and weekly averages for each hour for the full year of 2024 (52×24). Data ranging 2018-2023 is used to produce: (i) hourly spot price forecasts for each of the $7 \times 24 = 168$ delivery periods in the first week of 2024; and (ii) forecasts of the weekly average hourly spot price for each of the $52 \times 24 = 1,248$ hour-of-week cells across the full grid of 2024. All forecasts are constructed using only information available as of 31 December 2023, reflecting the information sets.

Motivated by literature and theory, we employ 2 approaches, one for each targetted task. For the first task, forecasting the first week of 2024, we use a panel model, LSTM neural network and ensemble. On the other hand, for the full year of average hourly day-ahead price forecasts, we use a calendar model. Variable selection and feature engineering forms a key part of the general modelling approach, beyond model architecture choice in this project, motivated by results from the literature and findings in exploratory data analysis.

We find that our forecast models generate many favourable characteristics of a forecast, capturing general patterns and dynamics well, across multiple days of the week (for i) and weeks of the year (ii). For the first task, local errors due to unforeseeable shocks manifest which the LSTM is better suited to but we find that a partial ensemble generates the best results, exploiting the LSTM's short memory gains and the panel's smooth dynamics. For the second task, the calendar model, systematic errors are identified, likely caused by solar production neglected in this paper but produces strong general performance in other areas. Nonetheless, as per Ockham's razor, it is striking how comparable the performance of a simple heuristic approach (median) is relative to our calendar model.

Overview

Modern electricity markets are unusual: because grid-scale storage is not yet economically viable, electricity must be produced at the moment it is consumed, and supply is sourced from a combination of near-zero marginal cost renewables and high marginal-cost dispatchable thermal plants [1]. Characteristics of electricity markets, namely non-storability, demand inelasticity and the aggregation across varying marginal cost inputs, mean that price determination is complex and a non-trivial feature of electricity markets today. This, in turn, has given rise to sophisticated methods and means of forecasting electricity prices. Pronounced intraday and weekly seasonality, large tails in distributions and strong regime-dependence and structural breaks are all key considerations for electricity price modelling and are in violation of standard time series assumptions [2].

The day-ahead spot market is the primary venue for electricity price formation in Europe where, within the Nordpool, buy and sell orders are submitted by 12:00 CET and a comprehensive market clearing auction simultaneously determines equilibrium prices for every single hour of the subsequent day across all bidding

zones. By simultaneously determining the full vector of the 24 hours of prices, prices across delivery hours are jointly set and predictable, conditional on the information set of the prior day. In particular, hour-of-day prices cannot be predicted from earlier hours of the same day, because they are set simultaneously rather than sequentially. Econometrically, this motivates a multivariate modelling approach over the 24-hour price vector rather than a collection of separate univariate models [3].

West Denmark, Jutland and Funen, acts as a rich natural laboratory for which to forecast electricity prices. DK1 is among the most renewable-intensive markets in Europe with wind penetration at almost half (49%), resulting in a highly non-linear, threshold-dependent relationship between wind output and price, as the literature documents [4, 5]. Given that DK1 is also part of the wider Nordic system with physical transmission links to Norway, Sweden, the Netherlands, Germany, the UK and DK2 (Zealand), cross-border flows present a further challenge in forecasting, given the means for electricity supply to be imported and exported. Indeed, the DK1-Germany interconnector and German day-ahead prices are key predictors of DK1's hourly day-ahead or spot prices [6, 7] and are highly non-trivial.

Complicating the forecasting challenge further, Europe's 2021-2022 energy crisis characterised by high natural gas prices following post-COVID demand and the invasion of Ukraine means that elevated TTF gas prices (which are a key driver of electricity prices) could pose distributional limitations and be problematic for models trained on pre-crisis data [8, 9].

In all, electricity price forecasting is a challenge, beyond academic exercise, that is important for many economic agents. With EPEX spot market trading volumes reaching 868 TWh [10], accurate price forecasting is highly pertinent for arbitraging market agents, firms with major energy cost exposures and policy makers on a broader scale. Short-horizon and hourly/intraday forecasts are especially important for energy traders who optimise positions in day-ahead auctions and balancing markets. Longer-horizon forecasts of weekly and, by extension, monthly average prices are also key components of future contract valuation and managing multi-year price risk in power purchase agreements and larger-scale renewable investments [1, 11].

Information requirements and the accompanying issue of the subset of variables selected, along with the feature engineered space of variable inputs, and, finally, the model architecture are key considerations for forecasting at varying horizons. Variable selection is discussed more thoroughly, however, for instance, whilst day-ahead wind and load forecasts are key drivers for an hourly horizon, commodity prices and seasonal patterns become more important as high-frequency fundamentals lose predictability at longer horizon [12, 13].

II. DATA

Electricity, climate, energy and macro data was provided [1]. Electricity prices are the hourly spot (day-ahead) prices for the DK1 price zone and is the target which we forecast, with all 24 hourly prices being determined simultaneously at the noon auction the day before as aforementioned. Loads (effectively, electricity demand) are the hourly actual total demand in MW with the day-ahead forecast also being provided, available the day before delivery and thus able to be used in forecasting. Renewable production includes hourly data and current or intra-day estimates but are contemporaneously unavailable for forecasts as they are only available on the day of delivery. Similarly, transmission flows between neighbouring price zones are provided. Energy prices for natural gas, oil, coal and EU carbon emission allowances (hereafter, EUA) are available with close data used. Then, macro-financial series on the VIX, EUROSTOXX50, EURUSD, Eurozone GDP and Danish GDP/CPI are available at mixed frequencies. Finally, climate data, daily and at the station-level, sourced for the European Climate Assessment & Dataset (ECA&D) assigned to the DK1 bidding zone by station coordinates are used, spanning temperature, precipitation, sunshine duration and mean wind speed.

An external calendar data source was used: Danish bank holidays were sourced as an additional binary variable to further explain seasonality.

The data preprocessing system involved:

- **Date scope restriction.** All source files were filtered to be within 2018-01-01 and 2024-12-31 range, with models being trained only on data from 2018-01-01 to 2023-12-31. Data outside this window was dropped.
- **Timestamp normalisation.** The source files use a mix of time conventions: prices are already in UTC, while loads, renewables, and transmission use CET/CEST. All hourly series were converted to a single UTC timeline using the Europe/Copenhagen timezone rules, which correctly handles daylight-saving transitions.
- **Daylight-saving duplicate handling.** On the autumn DST fall-back day, the source Loads file contains two rows for local 02:00 (one in CEST, one in CET) that both map to the same UTC hour. These duplicates were collapsed to one row by averaging.
- **Normalising missing-data.** The source files use a variety of conventions for missing data, for example 'n/e' in transmission, '-' and 'N/A' in loads and renewables. All numeric casts use `TRY_CAST` so that any unparseable string becomes `NULL`. A dedicated audit script verifies that no real numeric values are silently lost.
- **Dropping uninformative and leakage-prone columns.** Columns that were either constant (e.g. the price zone identifier, always `BZN|DK1`), low value given other columns (e.g. the open, high, low, change_pct for commodities given price was available), or explicitly flagged by the case document as unavailable prior to that date (the `intraday`, `current`, and `actual` renewable columns) were removed from the final dataset.
- **Pivoting renewables from long to wide format.** The source renewables file uses a long format with one row per hour per technology. This was pivoted to a wide format with one row per hour.
- **Combining all sources into a single hourly panel.** After the individual source tables had been cleaned, they were joined into a single wide table (`hourly_features`) with one row per hour of the Case A window. The hourly spine was constructed by generating every hour between the first and last observed price timestamps, guaranteeing gapless coverage even if an individual source table had missing hours. Each source table was then left-joined onto this spine on the appropriate key: hourly tables (prices, loads, renewables, transmission) joined directly on `ts_utc`; daily tables (commodities, equity indices, VIX, USD/EUR, climate) joined on the calendar date; monthly CPI series joined on the first day of the month; and quarterly GDP series joined on the first day of the quarter. The result is a single analysis-ready table that combines twenty source files into a unified hour-level panel with the day-ahead price as the target variable alongside all explanatory features.
- **Station-level climate aggregation.** The ECA&D climate files are at station-day granularity, with many stations reporting only a subset of days. For each of the four variables (temperature, precipitation, sunshine, wind speed), stations reporting fewer than 80% of days in the window were excluded to remove noise from short-lived feeds. The remaining stations' values were averaged to produce one daily value. Also converted from the ECA&D native tenths units to standard units (°C, mm, hours, m/s) for readability.
- **Forward-filling missing data at hourly level.** Daily commodity and financial series, monthly CPI series, and quarterly GDP series are all lower-frequency than the hourly target. These were joined onto the hourly level final dataset and forward-filled. For example, coal price on the Friday will be

the coal price for all hours over Friday and the following weekend.

- **Calendar flags for weekends and Danish public holidays.** Two boolean indicator columns, `is_weekend` and `is_holiday`, were added. Danish public holidays were obtained from the Python `holidays` library.
- **Automated validation of the cleaned data.** A validation script runs 41 assertions on the final feature table, covering row completeness (no missing hours, no duplicate timestamps, exact split-table sizes), range sanity (prices, loads, renewable capacities, climate values in native units), internal consistency (forecast-vs-actual correlations, per-border transmission flows summing to the total), timezone alignment (solar peak hour in midsummer, weekday load peak in daytime hours), and behavioural correlations (wind-price negative, load-price positive). All 41 checks pass.

A. Variable Selection & Feature Engineering

We conduct variable selection in line with existing literature on hourly day-ahead electricity price forecasting in the DK1 price zone as well as our exploratory data analysis. Exploratory Data Analysis (EDA) and literature analysis for variable selection resulted in several implications for our feature space. Feature selection is important to judiciously select the most relevant variables, minimise multicollinearity between explanatory features and to reduce the dimensionality of data by excluding less relevant variables. Since existing literature focuses on a range of testing periods and available features, we cannot simply use selected features from a single paper. In particular, the post-Covid energy crisis due to Russia’s invasion of Ukraine has led to a potential structural change in factors impacting electricity prices which can be seen through correlation plots before and after 2022. For example, during and after the crisis, natural gas prices and German electricity flows have become an increasingly dominant electricity price driver, as seen in Figure 6.

Liu *et al.* [6] apply a SHapley Additive exPlanations (SHAP) for feature selection across 19 potential features which measures the marginal contribution of the addition of each feature. They select the 7 most relevant variables for the prediction of the next 24 hour spot prices simultaneously, testing predictions on August 2019 since it was a period of stable electricity prices in the DK1 region. Electricity price lags at the 1 day and 7 day level in both DK1 and DK2, intra-Danish electricity exchanges as well as cross border flows between DK1 and Germany are determined to be the most important variables. This matches the findings of Li and Becker [7] who highlight the dominant role of Germany in DK1 pricing. They test 62 features across 5 feature selection methods in the LSTM architecture, emphasising the sensitivity of feature selection for LSTM model results. Hence, we include 1-7 day lags of DK1 electricity prices in our baseline model and include net electricity exchanges with DK2 and Germany. We also sum net electricity exchanges with all other countries to obtain a third feature. Although we expect hourly lagged prices to be most significant at the 1-day and 7-day lag due to strong day of the week effects, we include all hourly lagged prices from 1-7 days in our baseline OLS model to explicitly model the temporal relationship of electricity prices. Our LSTM model instead accounts for temporal relationships through its long term memory state, without explicitly including lagged variables.

Due to the high renewable energy penetration in the Danish electricity market, wind production, which accumulates to 49% of Danish electricity production (Becker and Blatt [5]) is an important supply side feature with a negative correlation with electricity prices. Jónsson *et al.* [4] find that accounting for non linear effects of wind power significantly improves DK1 spot electricity price predictions which highlights the importance using the LSTM model which accounts for non-linear relationships in the neural network without specifying a non-linear functional form for wind production.

Becker and Blatt [5] conduct feature selection by selecting four months across different seasons and testing the significance of all exploratory variables in each month. They conclude that solar power is not significant

in any of the four months and due to the high seasonality of solar power which is incorporated in monthly dummies as well as its high hourly volatility which is susceptible to weather changes, we do not include solar power in analysis.

To account for strong across-year seasonal electricity demand, we include month and weekend dummies as well as public holiday dummies. These are also significant since these are the only exogenous variables which are known about January 2024 at the time of forecasting. Intraday seasonality is dealt with through having separate model coefficients for each hour of the day (although exogenous variables coefficients are shared across the hours).

Karabiber and Xydis [14] use backwards feature eliminations beginning with 7 regressors to measure the change in RMSE and MAE as regressors are removed from prediction. They conclude that removing temperature improves RMSE. In our EDA, we compute correlation coefficients (figure 13) and scatter plots of weather features with forecast load, concluding the strong linear relationship between them. This corroborates the findings of Karabiber and Xydis [14] as there is little exogenous contribution of weather features for prediction beyond the effect of weather on electricity demand. Karabiber and Xydis [14] also find a linear relationship between temperature and electricity prices specifically below 0°C which is suggestive of a dummy variable for temperature below freezing. This may be because temperatures below freezing have an additional effect on electricity prices through the wind supply side channel as it can cause icing on blades, deteriorating wind production efficiency. However, we find little linear correlation between temperatures below freezing and electricity prices (figure 1), hence we do not use the temperature dummy variable. As such, we do not include any weather related features such as temperature, wind speed and sunshine, since these affect electricity prices through either the demand side which is accounted for in predicted load or supply side which is accounted for in wind production and commodity prices.

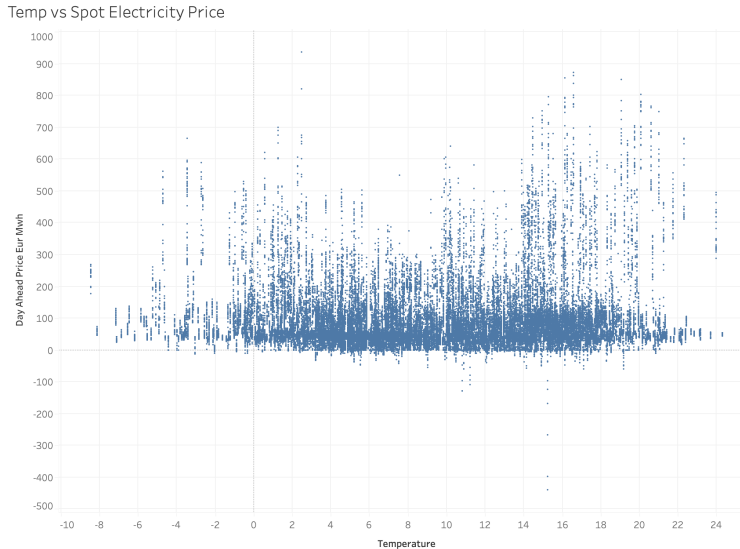


FIG. 1: Scatter plot of temperature and spot electricity prices

For the long term model, we must instead consider which variables are most conducive for prediction over much longer horizons than one week. In particular, any exogenous variables used must be sufficiently stable to be first predicted (Becker and Blatt [5]), and then used in forecasting across longer dates.

As such, we first include time dummies (weekend, month and public holiday) to account the strong seasonality of the price predictions. As for exogenous variables, we include a set of features that are indeed stable across years, and we make their 2024 prediction using averages from their values in previous years. We choose wind production, load, total net imports and weather variables (temperature, wind speed and

sunshine). We do not include commodity prices since these variables trend overtime and are very volatile at high frequencies. Hence, weather is an important variable to account for electricity supply, since intuitively, low wind speed which reduces wind production can lead to a higher demand and price of fossil fuels, driving higher electricity prices.

B. Exploratory Data Analysis

EDA reveals several key features of the target and explanatory variables, such as volatility, price span, periodicity, seasonality and regime shifts. We find patterns not emphasised in the existing DK1 forecasting literature. An examination of all variables was undertaken before variable selection but in line with the literature and our findings, the most important variables are maintained and communicated here in this order to reveal the key variables of focus.

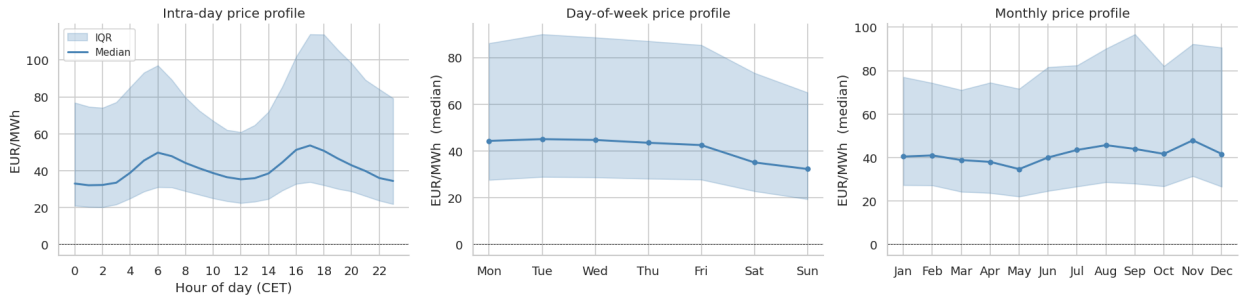


FIG. 2: Intraday, Weekly and Monthly Seasonality



FIG. 3: Holiday vs Non-Holiday and Weekend vs Weekday Intraday Seasonality

Primarily, there are 3 key seasonalities to consider in the target and amongst many covariates: intraday patterns, patterns across the week and seasonality patterns across the year. Figure 2 illustrates seasonalities and volatility (through the IQR) that arise intraday with morning and evening peaks coinciding with patterns of electricity use expected during workdays when people awaken and return home. Weekdays did not illustrate any particular difference but weekends and holidays are visibly different and, figure 3 also compares the intraday patterns of weekends, with noticeably smoothed morning peaks and energy prices lower all over. Across the year, there is evidence for price positive seasonality in the winter, with some change in summer (perhaps, for example, due to air conditioner usage). These patterns are also visible on autocorrelation plots which have clear peaks at 24 hours (1 day) and near 168 hours (1 week), figure 4.

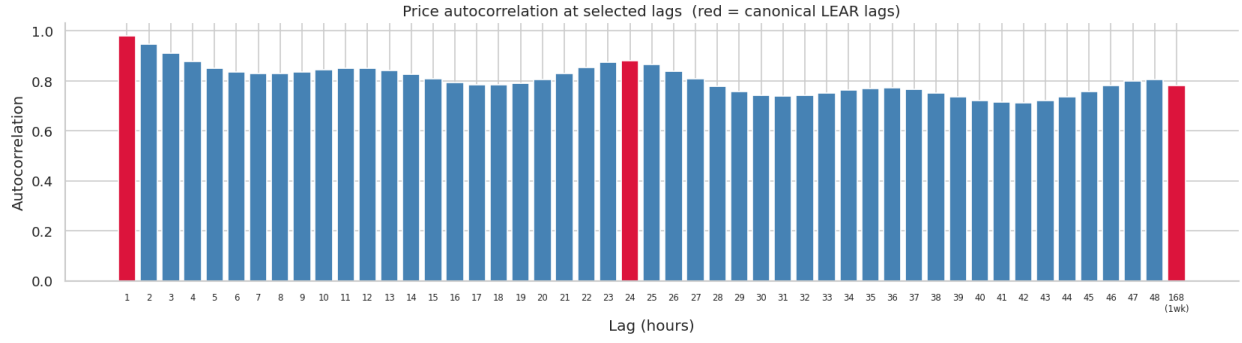


FIG. 4: Autocorrelations of DK1 Energy Prices

A heatmap of the median (removing outliers) 5 electricity price over each average hour for each month. A vague doughnut shape reveals a pattern of morning and late afternoon price rises but summer afternoon price cheapening, potentially reflecting earlier described patterns.

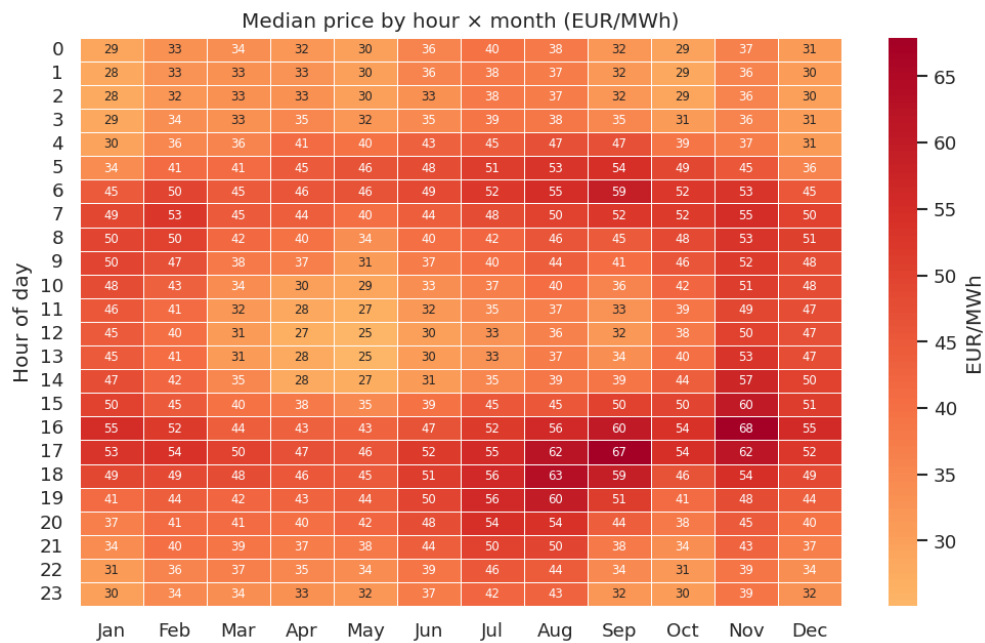


FIG. 5: Heatmap of Median DK1 Energy Prices

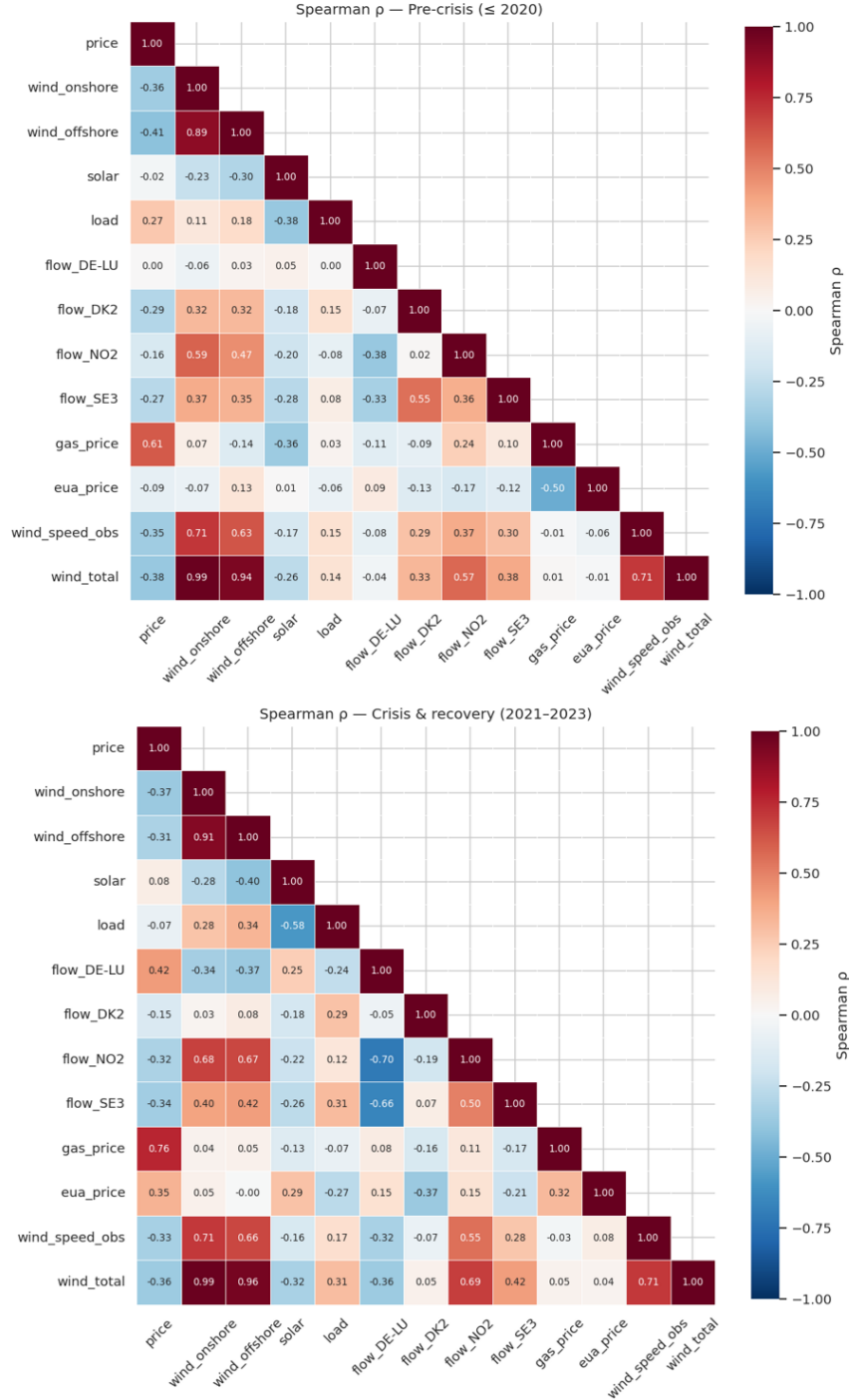


FIG. 6: Correlations Pre and Post 2021

When considering feature importance, a correlation matrix provides insights into the variables of greatest importance. However, we also consider how regime-shifts may result in a changing underlying hierarchy of feature importance, as discussed in our variable selection. Figure 6 illustrates a changing correlation structure in the data. We use this to establish which variables are highly collinear, in variable selection and to understand feature importance to support variable selection. Figure 7, for example, is pertinent in illustrating how post 2021, TTF gas prices became a much stronger predictor of DK1 electricity prices (also visible in the correlation matrix).

The patterns in renewable production over time, figure 8, are also pertinent to understanding the nature of

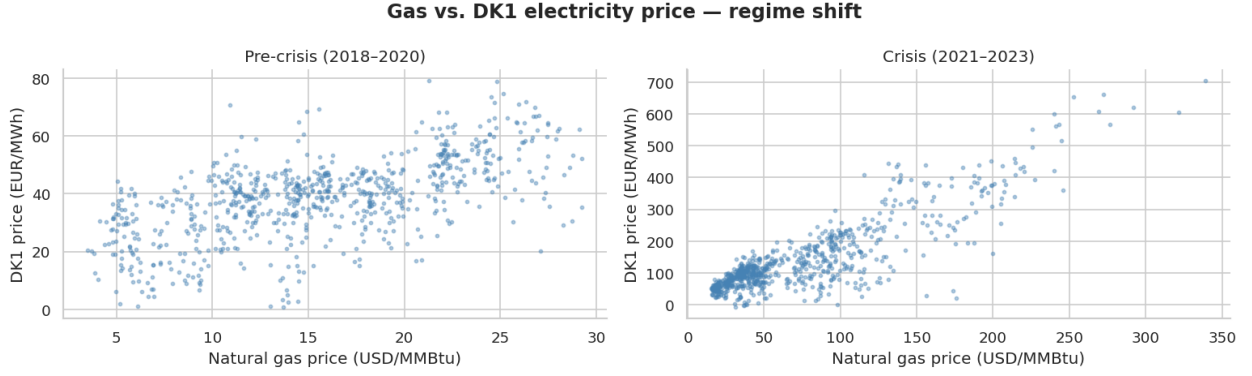


FIG. 7: Gas and Electricity Relationship Pre and Post 2021

the problem. At the monthly level, solar generation follows a clear and predictable seasonal cycle, peaking in the summer months and producing very little in the winter. This monthly regularity, however, masks substantial day-to-day and hour-to-hour variability, which makes solar much harder to forecast at the daily resolution relevant for the day-ahead auction. The figure also shows the clear growth in installed solar capacity over the sample period.

Wind generation is and remains an important source of electricity in DK1, with a baseline contribution visibly in line with findings of the literature. Unlike solar, wind exhibits no strong seasonal cycle: figure 8 shows monthly wind penetration oscillating around a roughly stable mean across the full sample, with winter months no less productive than summer ones. This is a structural advantage for the Danish system, since the wind contribution to the supply mix does not collapse in any particular season. However, wind is still volatile at the daily level, which poses a challenge for short-horizon electricity price forecasting. Residual demand that cannot be met by renewables is provided by dispatchable thermal power plants operating at a higher marginal cost, so hours with low renewable output translate directly into higher equilibrium prices in the day-ahead auction.

Finally, in the Appendix, figure 13 illustrates the interaction between several covariates with load and the target, plotted in 2-dimensions. This provides insights into the qualitative direction of each relationship but also the potential nonlinearities. Climate variables and the target reveal a simultaneous causality issue regarding variables and load but also provide insights into the broader interaction of climate conditions and electricity supply, through the indirect connection on the load, which supply must act to clear. It is also striking that, in contrast to the literature, for instance, the German flows of electricity appear rather insignificant in electricity price impact. This is related to threshold non-linearity issues caused by wind production in particular, discussed throughout this paper (in the variable selection and methodology overview), whereby wind production can mean that German flows of electricity only have an effect at certain levels of wind electricity production.

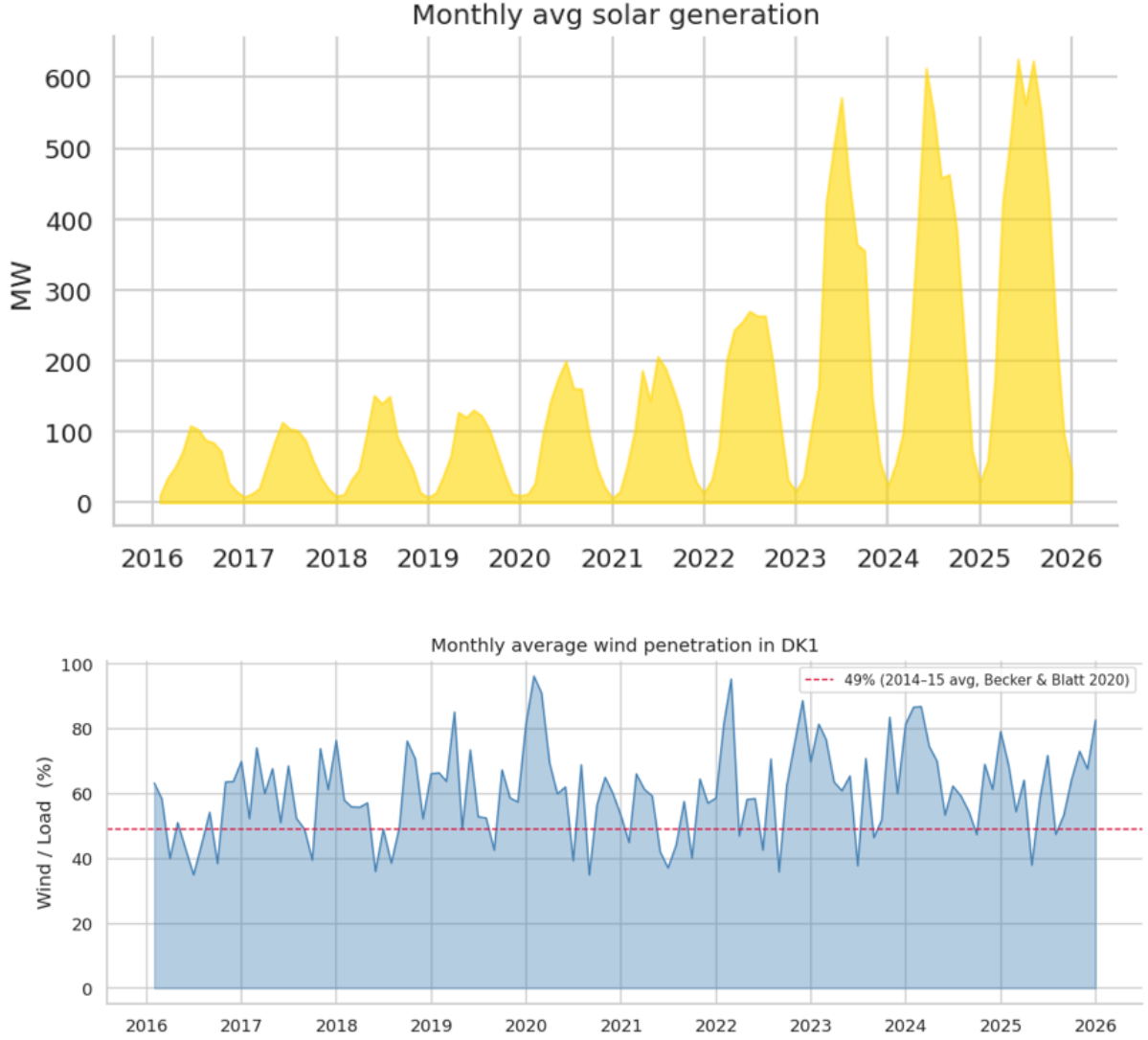


FIG. 8: Renewable Production over Time

III. METHODOLOGY

A. Methodology Overview

The methodologies employed in electricity price forecasting are broad and have been evolving rapidly with data availability and the advent of machine learning (ML), developing into a substantial field of its own, distinct from the broader macro/financial time-series forecasting. As explained in the overview, the physical and institutional idiosyncrasies of electricity markets have led to an emergent shift from simple heuristic results (i.e., taking the mean or median hourly day-ahead spot forecast) and parametric, simple autoregressive benchmarks, to rich hybrid architectures with modern interpretability tools which aim to retain econometric interpretability and economic intuition (see, e.g. [2]). We organise this overview of methodology in the electricity price forecasting literature around themes relevant to our modelling choices, later justified, following the evolution of forecasting methodologies from parametric benchmarks to deep learning and spatial models.

Incompatible with Gaussian assumptions underlying the classical time-series models, non-storability and

the need to balance electricity demand with supply, requiring marginal cost diverse energy inputs, results in heavy tails, positive skewness and transient spikes [2]. Merit order effects, due to the penetration of renewables like wind and solar which carry near-zero marginal costs, dispatched first, shift supply leftwards, reducing equilibrium prices [1]. This creates non-linear, threshold style relationships between wind production and electricity prices in the DK1 price zone [4, 5]. Jónsson *et al.* [4] demonstrate that explicit modelling of a nonlinear wind-price interaction, relative to the treatment of wind as a linear covariate, significantly improves DK1 day-ahead forecasts. Liu *et al.* [6] find that through SHAPley values that threshold effects exist at 800 MW on the DK1-Germany interconnector additionally, for instance. Ergo, nonlinearities in the DK1 price zone that are established motivate model architectures that can model complex input-output mappings more flexibly than parametric models where nonlinear functional forms have to be pre-specified. Further, many patterns of highly predictable daily, weekly and annual cycles of activity resulting from human and natural process, impart strong multi-frequency seasonality to prices [14, 15].

The parametric, workhorse EPF benchmark was and remains the AR framework, augmented by LASSO regularisation as codified by Lago *et al.* [15]. Their LEAR (LASSO Estimated AutoRegressive) model has been employed with 240 regressors, including lagged prices at $d - 1$, $d - 2$, $d - 3$, and $d - 7$, generates strong forecast performance. Many more sophisticated structures, examined alongside this, underperform and, so, as a simple benchmark, parametric methods with some augmentation appear to provide a high bar for generating forecasts, with more complex architectures requiring strong performance gains in order to become considered valid. Uniejewski *et al.* [16] confirm that prices for the hourly day-ahead forecasts on the previous day and one week ago are the most consistently LASSO-selected regressors across Nordic markets, and that day-of-week dummies are significant, providing confirmation for employing LEAR models for EPF.

More firmly within the realm of parametric alternatives, SARIMA and SARIMAX models provide a natural means of modelling the multi-frequency seasonal structure of electricity prices. Karabiber and Xydis [14] apply ARIMA among other methods to DK1 price forecasting and find that, among all models, ARIMA yields the best mean error performance. Zhao *et al.* [17] further develop these findings that parametric benchmarks can be powerful, by showing that ARIMAX-GARCH specifications can improve short-term forecasts by jointly modelling conditional heteroscedasticity alongside the mean equation.

The limitations of parametric models in capturing nonlinearities, interactions among predictors and highly complex multi-scale temporal seasonalities alongside the advent of machine learning more generally, have motivated alternative approaches to EPF. The most prevalent of these is the LSTM (long short-term memory) model. Kılıç *et al.* [18] find that when applying LSTM networks to the DK1 intraday price forecasting problem, LSTM outperforms many benchmarks. The success of the performance of neural networks is rationalised by its ability to flexibly capture nonlinearities, due to its neural network architecture with nonlinear activation functions, and capture dependencies over long sequences, due to its gating mechanism, allowing it to selectively retain useful information and remove useless information over a long time.

Li and Becker [7] use different selection algorithms with LSTM architectures and find that no single feature selection rule is able to dominate across all market conditions. The implication is that the choice of selection algorithm affects LSTM accuracy non-trivially and that the market conditions and idiosyncrasies are pertinent to optimal variable selection for an LSTM. This is relevant to the careful, theoretically and empirically grounded variable selection and feature engineering employed in this project which seeks to develop superior forecasts with an LSTM architecture, through the initial filtering of variables.

In a similar vein, Maciejowska *et al.* [19] demonstrates that improving the quality of exogenous inputs is a key avenue for further performance improvements. The quality of day-ahead forecasts for different variable inputs is a key factor for model performance and, suggests that variable selection and feature engineering, rather than model selection, is important for developing further, improved forecasts. Uniejewski and Ziel [20] support this further by using probabilistic forecasts to also demonstrate that inputs can reduce price

forecast error (if point forecasts are not the only forecast being generated). The NBEATSx architecture of Olivares *et al.* [21] decomposes trend, seasonality and exogenous input for price forecasts, verifying the importance of forecasted inputs into a model for accurate EPF.

The use of machine learning has also been varied in more recent and alternative strand of the literature where it explicitly models the spatial dependence structure among interconnected bidding zones. Yang *et al.* [22] transform energy price data of multiple countries, and their flows, into a graph, applying a STGNN (spatio-temporal graph neural network) for EPF, finding strong performance also.

Combining predictions from different models has also been explored and provides gains, as per the ensemble principle. Karabiber and Xydis [14] is able to demonstrate using ARIMA forecasts and others that a model average provides performance which is better than any individual model, effectively illustrating the ensemble principle. When Ziel and Weron [3] test various LASSO regression variants across different datasets including DK1, they find a role for forecast combination of univariate and multivariate approaches to generate superior results. More sophisticated ensembles, explored by Madadkhani and Ikonnikova [8] use a random forecast, XGBoost and Bayesian Additive Regression Trees (BART) for German day-ahead prices and also find that ensembles succeed in most predictions. Where they do fail is during the regime-shocks in 2021 where parametric specifications are more robust, demonstrating the maintained importance of parametric benchmarks in our research.

Hence, a thorough examination of the literature on EPF methods finds that both parametric and ML methods are important in the forecasting of electricity prices. Indeed, the family of ARIMAX models on the parametric spectrum and ML models, namely LSTMs, provide strong predictions. Our contributions follow the findings of literature in that careful variable selection is where most gains yet can be developed, whilst retaining model architectures that have proven successful.

B. Task 1a: Panel Regression

To forecast all hours of the first week of January 2024 electricity prices, we model 7 panel regressions $\tau \in \{1, \dots, 7\}$, with a separate model for each forecast day. For hour h in a τ day ahead forecast, the panel regression is specified as follows:

$$p_{t+\tau,h} = \alpha_h + \sum_{j=0}^6 \beta_j p_{t-j,h} + \gamma' X_{t,h} + d_t + m_t + b_t + \varepsilon_{t,h}, \quad (1)$$

Where $p_{t+\tau,h}$ denotes the spot electricity price τ days ahead at hour h . The term α_h represents hour fixed effects. The terms $\beta_j p_{t-j,h}$ are the j -day lagged electricity prices, capturing temporal dependence through historical prices. The vector $X_{t,h}$ includes exogenous controls at time t and hour h , namely wind production, net flows from DK1, flows from Germany, flows from other countries, commodity prices, and EU carbon credit prices. The variable d_t denotes weekend fixed effects, m_t represents month fixed effects and b_t represents public holiday fixed effects.

Since this is a parametric model, we must explicitly account for non-linearities in features such as wind production which we choose to keep linear in the baseline model. The LSTM model accounts for these non-linearities without specifying a parametric form, allowing us to analyse the potential gains for non linearity when comparing the panel regression with the LSTM model.

C. Task 1b: LSTM Architecture

The LSTM is well suited to Task 1 because its gating mechanism excels at exploiting the high-frequency, recently-observed dynamics that dominate short-horizon price formation.

Because the 24 hourly prices of delivery day d are cleared jointly at the noon auction on day $d-1$, they cannot be forecast recursively from earlier hours of the same day. We therefore cast the problem as a multivariate sequence-to-vector task: the target is the full price vector $\mathbf{p}_{t+\tau} = (p_{d,1}, \dots, p_{d,24}) \in \mathbb{R}^{24}$, predicted in one shot from the information set $\mathcal{I}_t$, following Ziel and Weron [3] and Li and Becker [7]. At each historical day t we construct an input sequence of length $L = 28$ in which each step stacks the 24 hourly prices together with day-ahead load and wind forecasts, net transmission flows with Germany, DK2 and the remaining interconnectors, the TTF gas and EUA carbon closes, and calendar indicators. Lagged prices enter up to 7 days, consistent with the 1st, 2nd, and 7th lags repeatedly found dominant in Nordic EPF [15, 16]. Continuous features are standardised using only training-sample statistics to prevent look-ahead leakage.

The LSTM has one layer followed by a dense output head. At each step, the cell updates its hidden state $\mathbf{h}_s$ and cell state $\mathbf{C}_s$ through the standard gating recursions

$$\mathbf{f}_s = \sigma(W_f[\mathbf{h}_{s-1}, \mathbf{x}_s] + \mathbf{b}_f), \quad \mathbf{i}_s = \sigma(W_i[\mathbf{h}_{s-1}, \mathbf{x}_s] + \mathbf{b}_i), \quad (2)$$

$$\mathbf{o}_s = \sigma(W_o[\mathbf{h}_{s-1}, \mathbf{x}_s] + \mathbf{b}_o), \quad \tilde{\mathbf{C}}_s = \tanh(W_C[\mathbf{h}_{s-1}, \mathbf{x}_s] + \mathbf{b}_C), \quad (3)$$

$$\mathbf{C}_s = \mathbf{f}_s \odot \mathbf{C}_{s-1} + \mathbf{i}_s \odot \tilde{\mathbf{C}}_s, \quad \mathbf{h}_s = \mathbf{o}_s \odot \tanh(\mathbf{C}_s), \quad (4)$$

where $\sigma(\cdot)$ is the logistic sigmoid and $\odot$ denotes element-wise multiplication. The additive cell-state update permits gradient flow across long sequences, letting the network retain weekly and seasonal regularities while remaining responsive to recent shocks. The final hidden state is passed through a linear layer that maps directly to $\hat{\mathbf{p}}_{t+\tau} \in \mathbb{R}^{24}$, so one forward pass delivers a complete τ -day-ahead forecast. The architecture is attractive for DK1 because it can learn, without hand-specified functional forms, the threshold wind-price interaction [4, 5] and the non-linear dependence on the DK1-Germany interconnector [6].

Hyperparameters (L , number of layers, hidden dimension, dropout, learning rate, batch size) were set at default levels. We train by minimising MSE using Adam with early stopping on a 90-day validation set at the end of the period, and the model was retrained on the full training set with the selected best number of epochs.

D. Task 2: Calendar Regression

To forecast hourly spot prices for every week in 2024, we conduct the following, so-called 'calendar' regression for a τ day ahead horizon $\tau \in \{1, \dots, 365\}$ for hour h :

$$p_{t+\tau,h} = \alpha_h + d_t + m_t + b_t + \gamma' X_{d,h,m,b} + \varepsilon_{t,h}. \quad (5)$$

Where $p_{t+\tau,h}$ is the spot electricity price τ days ahead at hour h . The variable d_t represents weekend fixed effects, m_t denotes month fixed effects, and b_t captures public holiday fixed effects, and the inclusion of these numerous date dummies give this model its name.

We also include a set of exogenous variables, $X_{d,h,m,b}$, as discussed in section II, which are wind production,

load, total net imports, temperature, wind speed, and sunshine. These variables take their actual values during training, but must be predicted for the out-of-sample forecasts for 2024. As such, we take historical averages of these variables for the same hour h , month m_t , and day type d_t and b_t (weekday and holiday) in the training data as a prediction for their value in 2024. After the hourly predictions for the entirety of 2024 are generated, the mean prediction for each week and hour cell is taken as the final prediction for the long-term prediction task.

IV. RESULTS, DISCUSSION & ROBUSTNESS

A. Task 1: Panel, LSTM and Ensemble

The results in figure 9 depict the actual hourly day-ahead DK1 prices for the first week of 2024 and both methods: (a) the panel regression and (b) the LSTM, along with an ensemble of both. An alternative view of the same results, with one panel per hour of the day rather than per day, is provided in the appendix (Figure 14).

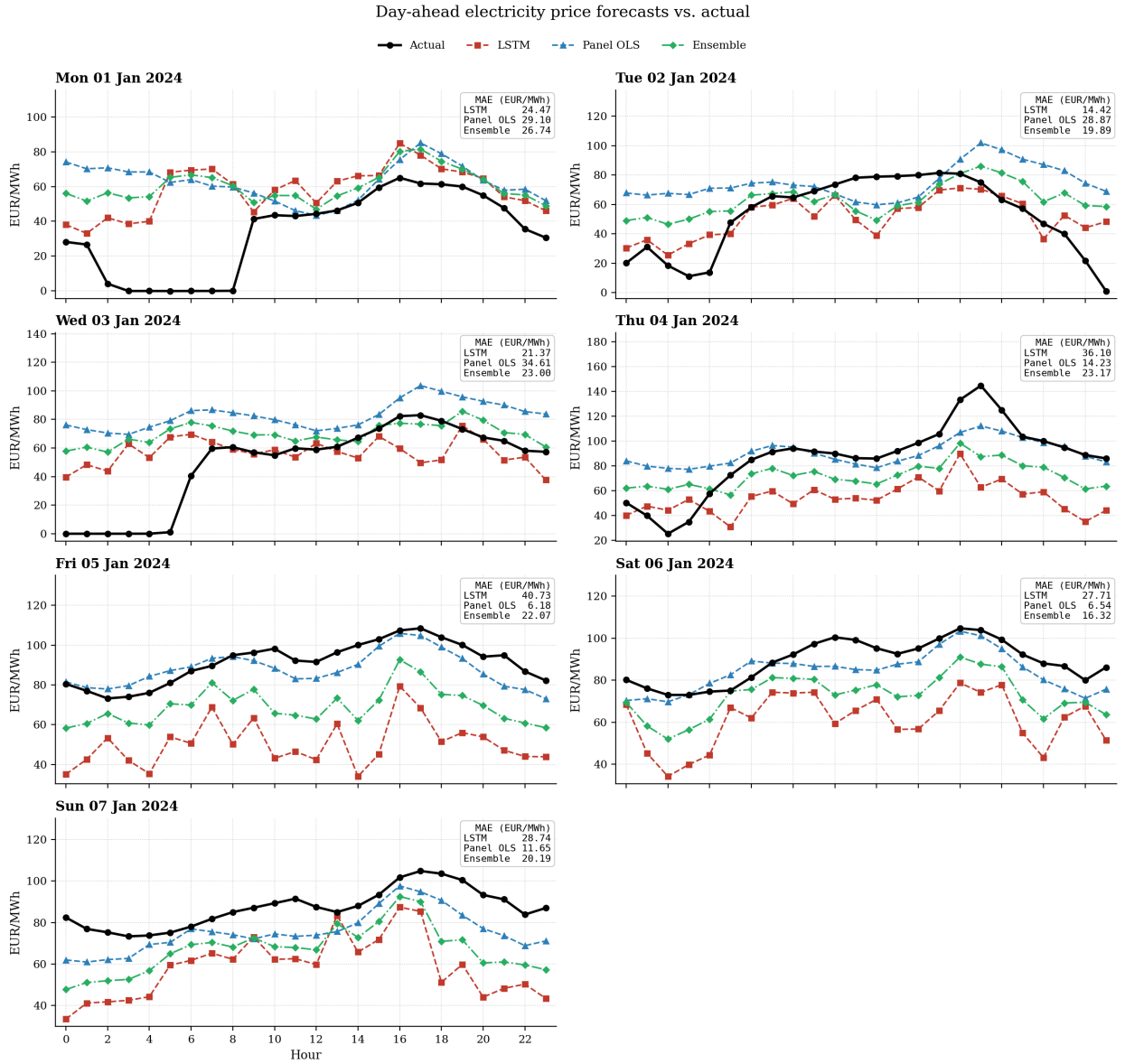


FIG. 9: Result for Task 1, split by day

The panel regression is clearly able to identify a general pattern which, on average, appears reliable but can be locally completely off. In the first few hours of day 1, for instance, when energy prices are near zero and negative, the panel regression which models longer-term trends entirely misses the mark and produces poor forecasts, with large errors, driven by the local failures. At different points throughout the day, however, the model appears successful.

This is entirely different for the LSTM. It captures, based on its short term memory, patterns and correctly identifies a fall in the first few hours of the first week of 2024, producing a significantly improved forecast. The errors are significantly lower. Generally, it is visibly apparent that the forecasts of the LSTM are less smooth and jump around which is likely driven by the short term memory retaining recent shocks/noise embedded in the forecasts as well as the non-linearity of features. The performance of the LSTM is better than the panel for the first few day, however, it quickly deteriorates to become worse than the panel in the later days of the week. This can be rationalised by the short-term memory of the panel clearly fitting to the recent observations where prices were low in the second half of the week and, so, the LSTM generates forecasts that consistently undershoot actuals. Hence, the LSTM is strongest at forecasting subsequent days but not the entire week ahead.

When combining these both into an ensemble, the ensemble principle holds, as anticipated from literature. The ensemble exploits both the broader pattern recognition of the panel and the short term memory strengths of the LSTM to generate forecasts that are visibly smoother and less random, but also which produce lower errors across the days. However, it remains too skewed towards the LSTM in the later part of the week when the LSTM's errors worsen.

Across the week, the LSTM's MAE is 193.54, the panel's MAE is 131.18 and the ensemble has MAE of 151.38. If the ensemble is used for the first half of the week (Mon-Thurs) and panel for the second half (Fri-Sun), the partial ensemble's MAE is the lowest at 117.17.

Results by hour across each day are provided in the appendix for examination of whether the models are forecasting specific hours of the day particularly well / poorly.

B. Task 2: Calendar Regression

The results of the calendar regression for producing forecasts for the hourly day-ahead prices across the whole year (52×24) are provided in figures 10,11 & 12.

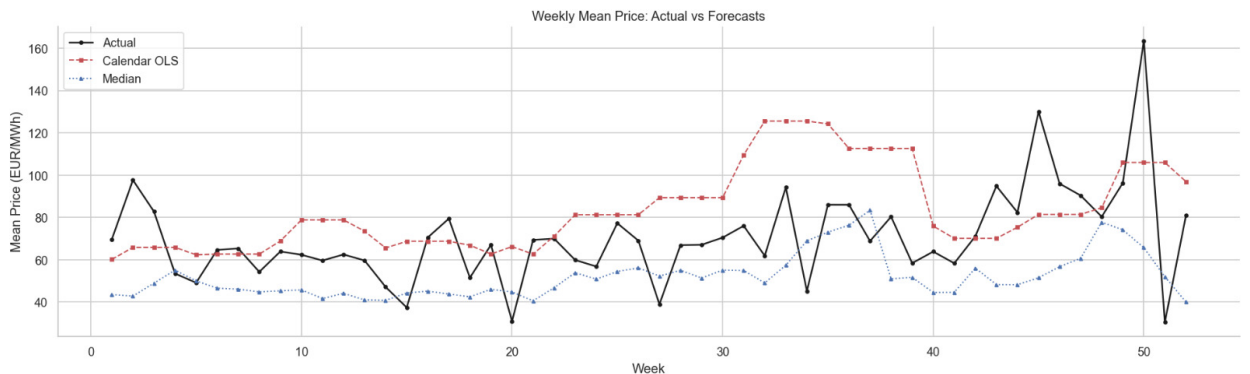


FIG. 10: Task 2 Results Across the Year

The calendar regression captures the bulk of patterns across the year based on historical activity and in many of the forecasts produced, the shape of the forecast curve reflects the same patterns. Differences are to be expected with the idiosyncrasies that arise in a given year; however, it is a testament to the importance of

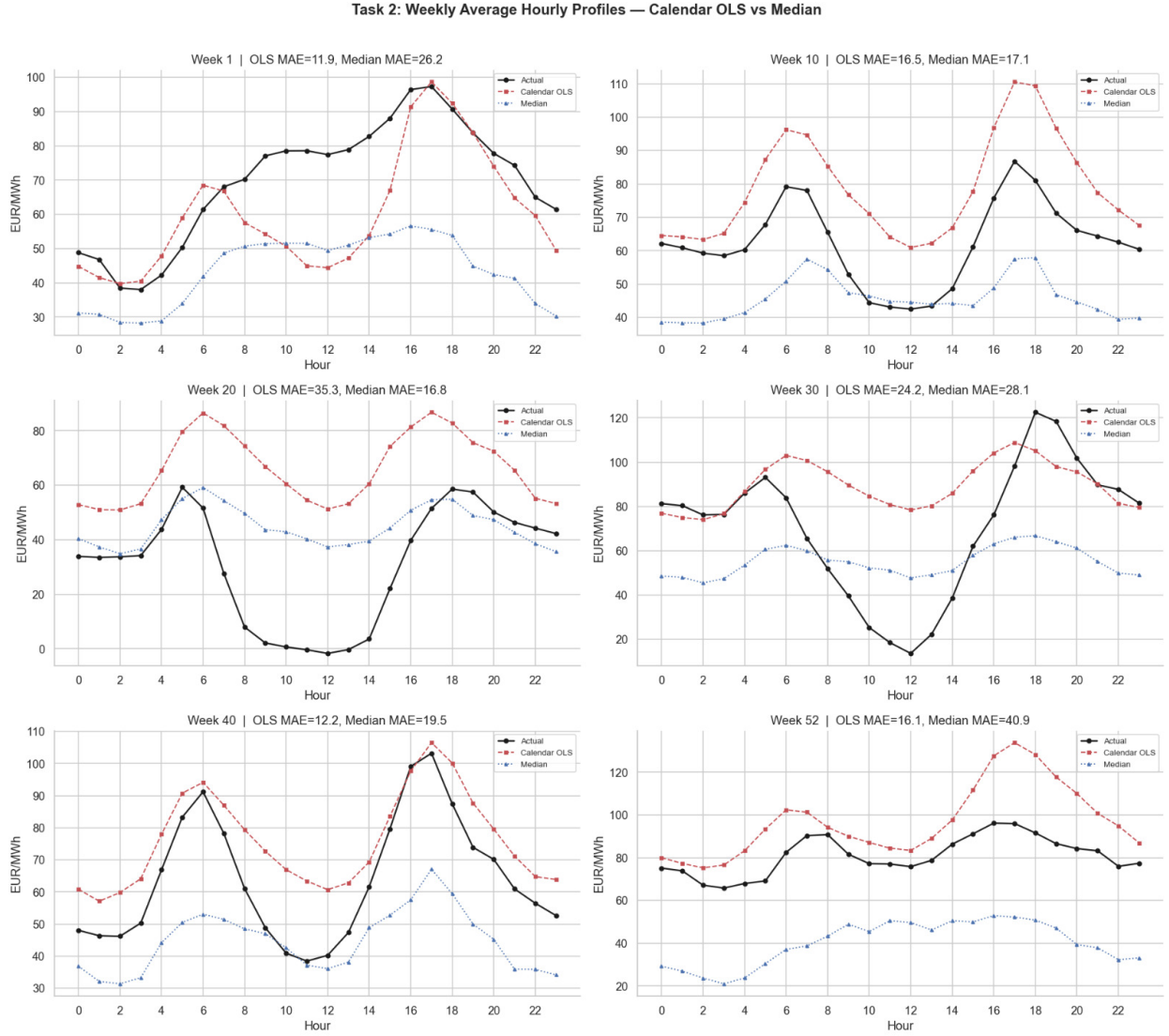


FIG. 11: Task 2 Results by Week

seasonality and predictability of electricity production from certain climate variables of how these patterns can be forecasted. It is also true that, this holds for many weeks across the year and that the performance does not degrade over time.

Perhaps most useful are the insights given by the heatmap of the forecast errors in figure 12. They illustrate that the forecast errors are often small. However, they highlight a key issue: a doughnut hole. Indeed, the results illustrate that the calendar model systematically overpredicts electricity prices. This is reminiscent of the doughnut ring found in the exploratory data analysis section and, is clearly related to summer-time occurrences. One potential explanation for this is exclusion of solar production from the model. Solar production is highly seasonal in the day-time during the summer weeks of the month and also increasing seemingly exponentially in production. If this is a continuing trend, increased solar production in the year of 2024 is likely to have contributed to falling marginal costs in the production of electricity and, therefore, reduced electricity prices. By neglecting this, performance is worsened on a systematic basis for summer noon hours. Whilst initially neglected due to trending behaviour, possible considerations are discussed in the extension V.

They also highlight certain weeks where performance is particularly poor. Week 51 is a week that is consistently overpredicted and week 50 is consistently underpredicted. This can be rationalised as the lead

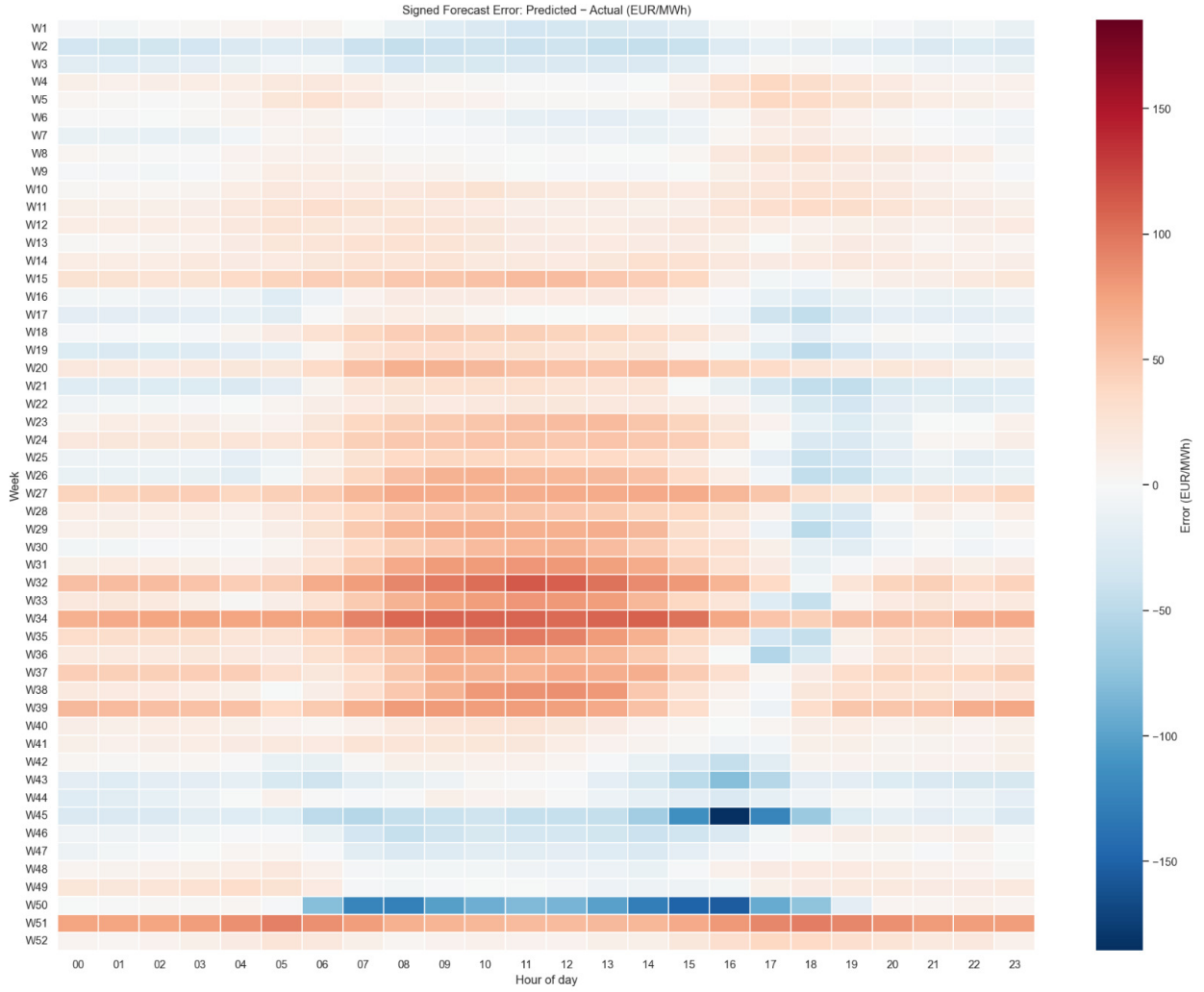


FIG. 12: Task 2 Results Heatmap

up to Christmas and economic activity causing a season price hike whilst Christmas eve, Christmas day and Boxing day inactivity causes a price fall. The implication here is that consideration of factors like holidays, for specific weeks, is pertinent accurate week-by-week forecasting. To a lesser degree, a similar pattern holds in week 15 which Easter holidays typically fall into.

As a benchmark the median provides a point of comparison. It appears to do well at getting a middle view across the year 10 but in many cases, across specific weeks 11, it is visually apparent that the calendar model is better modelling level and seasonal dynamics. The MAE across the year of the calendar model is 26.411 whilst the median is 26.583. This suggests that there are gains of the modelling conducted; however, as per Ockham's razor, often, it is striking how strong the performance of simple parametric methods or heuristics can be. The gains manifest in terms of the general pattern that the calendar model better replicates but the level differences and particular points of over and undershooting result is comparable MAE to the median, which tends to follow a significantly smoother forecast.

V. EXTENSIONS

A. Systematic Variable Selection

Although we utilise literature based and intuitive feature selection, a potential robustness check would be to compare our selected features with a systematic feature selection model such as Least Absolute Shrinkage and Selection Operator (LASSO) to select features with non-zero coefficients.

LASSO optimisation minimises a mean squared error (MSE) loss function subject to an l1 penalty and relies on the assumption of model sparsity which means that only a small subset of potential controls have a non-zero coefficient in the true data generating process. We can assume this to be true in electricity prices since we expect weather related variables to have low explanatory power after accounting for demand and supply side factors.

$$\min_{\beta} \frac{1}{n} \sum_{i=1}^n (y_i - x_i' \beta)^2 + \lambda \sum_{j=1}^p |\beta_j|$$

Alternatively, in line with the literature, (Liu *et al.* [6], Li and Becker [7]), use SHAP for feature selection which measures the marginal contribution of each additional feature.

B. Task 1 – Extension of ensemble with tree based models

We find that the ensemble generates smoother predictions which are less random and capture the strengths of both the panel regression and LSTM. Madadkhani and Ikonnikova [8] find when predicting day-ahead German electricity prices, ensemble models averaging outputs of Random Forests (RF), XGBoost (XGB) and Bayesian additive regression trees (BART) outperform individual base models in most years. Hence, a potential extension is to include tree based models such as a RF or XGB to our ensemble. RF are bagging models where several trees are trained in parallel and averaged whereas XGB are shallow trees which are trained iteratively by training on the residuals of previous trees and adding forecasts together with a shrinkage weight to form an ensemble. Tree based models are non-parametric and can learn the potential non-linear patterns of wind production as the LSTM does but may perform better than the LSTM over a long horizon as lagged prices are explicitly included.

C. Task 2 – Prediction of Exogenous Variables

We choose to model exogenous variables as a historical average of chosen exogenous variables for the same hour h , month m_t , and day type d_t and b_t (weekday and holiday).

An extension would be to consider other methods to model exogenous variables such as an Autoregressive (AR) model. However, this means we would only be able to include variables with strong persistence and it may lead to compounding errors overtime towards the end of 2024. Other methods could include an average of first differences of variables such as solar energy production which could be beneficial in reducing errors in the sunny hours of the summer period where errors are on average highest (figure 12).

Further, heat map in figure 12 demonstrates that model performance is systematically worse in weeks with cultural holidays such as week 15 which corresponds to Easter and week 51 which corresponds to Christmas. Including dummy variables for Easter and Christmas may help to improve model performance in these weeks.

VI. CONCLUSION

In conclusion, this paper has generated forecasts for the first week of January 2024 (7×24) and for the average hourly spot price for the DK1 price zone across all weeks of 2024 (52×24). According to literature and our exploratory data analysis findings, the variable set was limited to a subset of variables, excluding weather in task 1 and commodity prices in task 2, with other feature engineered variables designed for the modelling pipeline. A review of the methodology justified our model choices, using a panel, LSTM and ensemble in task 1 and calendar model in task 2. The short term memory of LSTMs and general ability to capture dynamics via a panel for task 1 and, the ability of the calendar model to track patterns for multivariate forecasting across an entire year, provided grounds for the use of these models. Variable inputs, rather than model architecture, it is noted from the literature, is crucial to developing further gains in EPF and, hence, the contribution of this paper is the employment of a novel subset of variables, motivated by findings on data analysis and literature.

The short term memory of the LSTM in task 1 proves beneficial in the first few days of the week but degrades in future days, overtaken in performance by the panel model. When a partial ensemble is applied, the performance improves beyond each individual model, exploiting the gains from both models. In task 2, the calendar model proves strong at modelling general patterns, despite some level differences. As a benchmark, comparing to the median reveals that the model adequately captures key seasonal dynamics and confirms that performance does not degrade over the year when forecasting. However, systematic biases appear to exist with regards to certain weeks of the year and noon hours of the summer months. These are rationalised by the neglect of solar production variables and specific holiday periods.

This paper provides key directions for future research on EPF in DK1 to consider, including a comparison to systematic variable selection, an extension to tree-based ensembles in task 1 and predictions of exogenous variables for input into task 2.

DECLARATIONS

Statement about the use of hardware and software. Local hardware and software is used with collaboration over GitHub.

Statement about use of AI. The autocomplete function of Copilot was employed during coding. Claude AI was also used to assist with coding and debugging the model, this includes some charts built in Python which are presented in the paper. AI was used to assist in the literature review and the formatting of BibTex references but was not involved in idea creation or any of the write-up in this paper. This work belongs to its authors.

Appendix A: Additional Figures

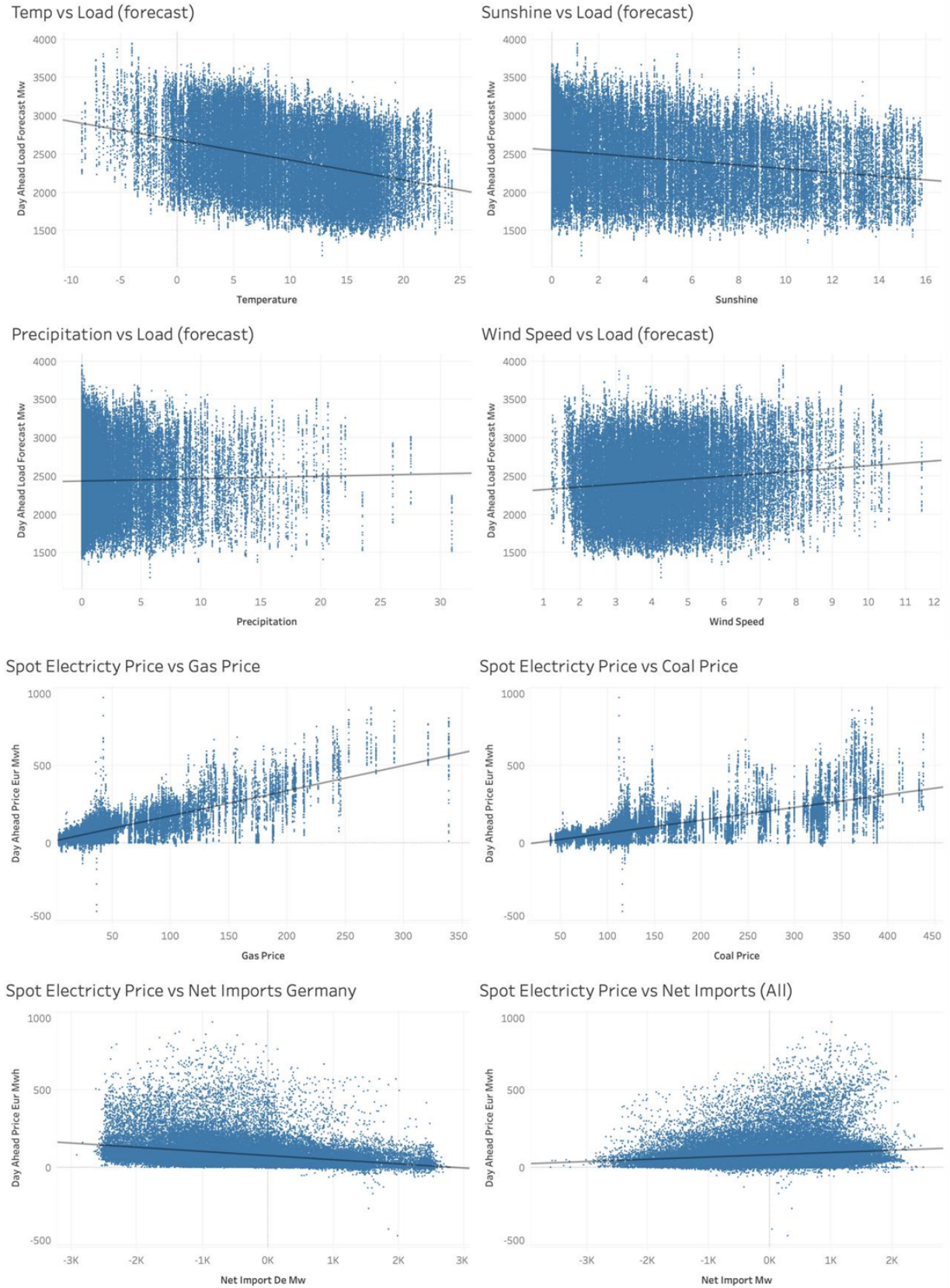


FIG. 13: Feature Interaction with Spot DK1 Price

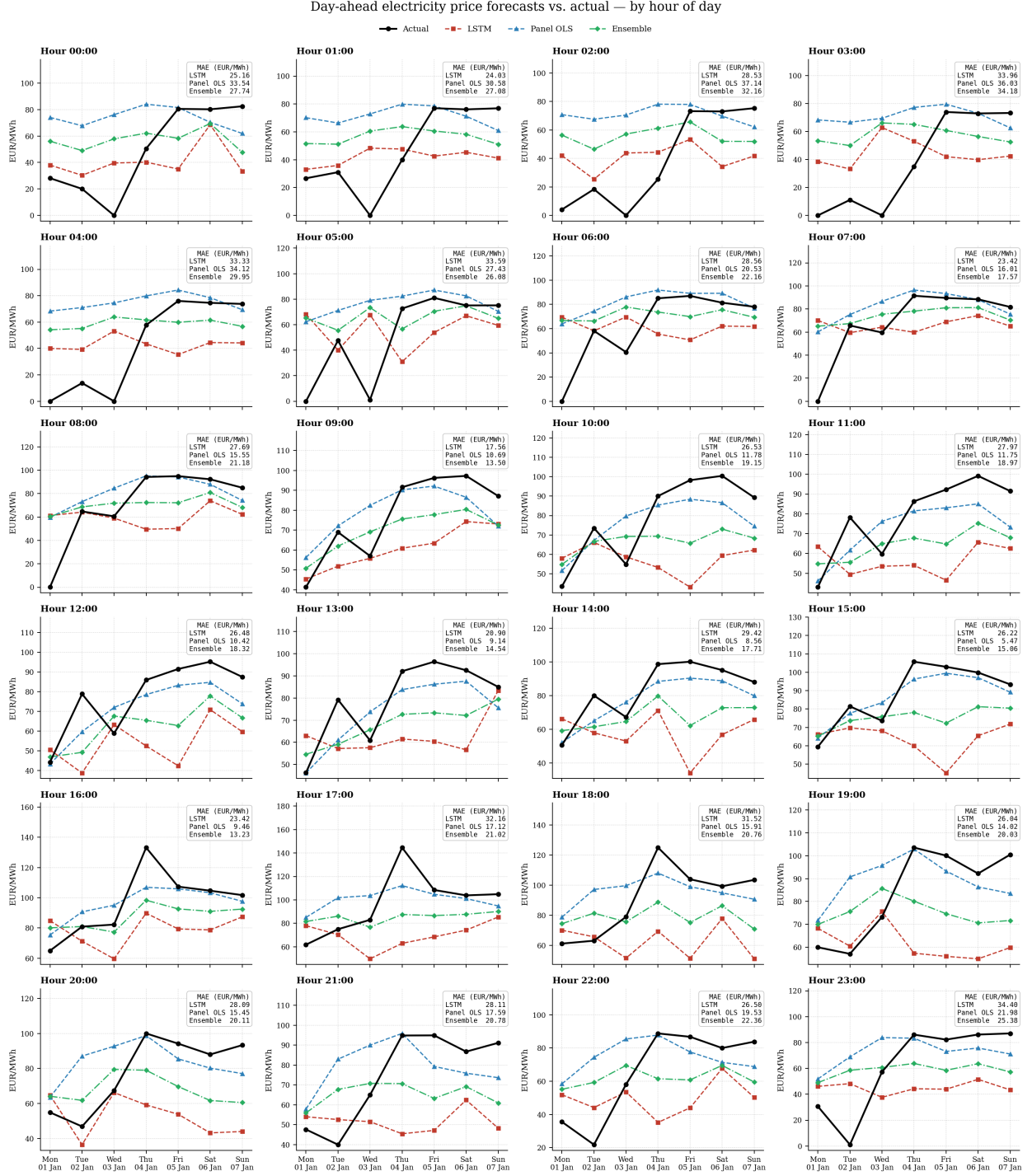


FIG. 14: Result for Task 1, split by hour

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